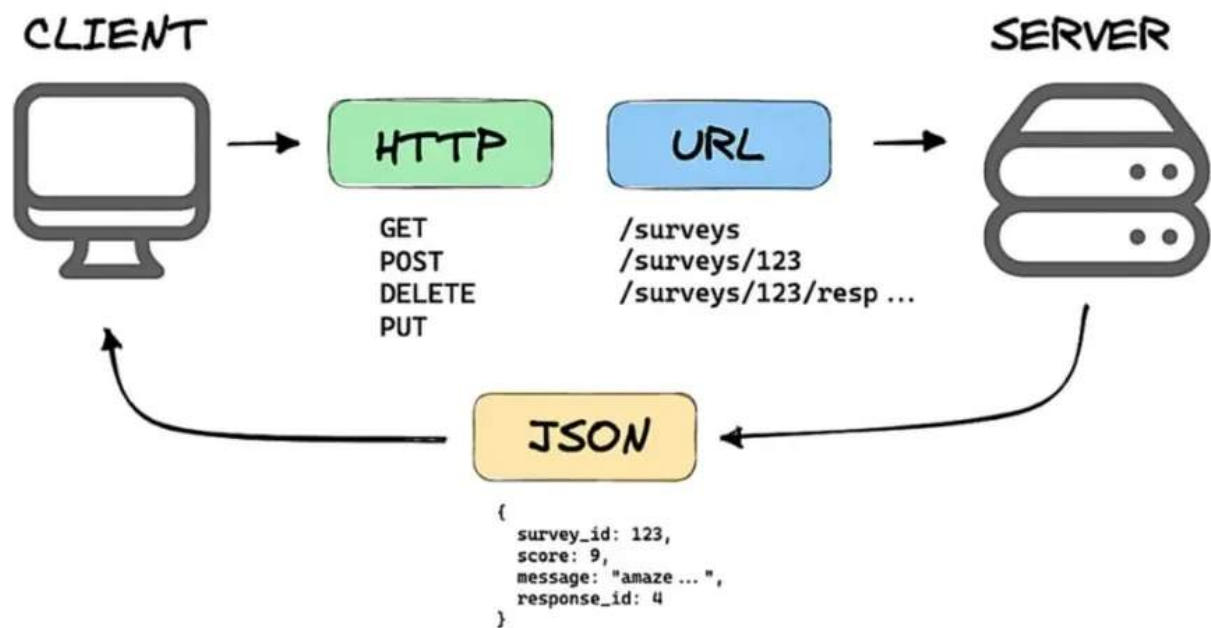
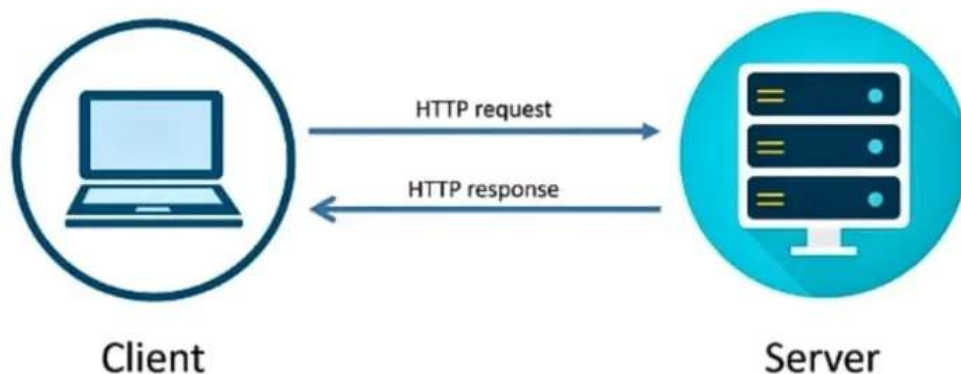


REST API Explained



Swipe>>>

A REST API (Representational State Transfer Application Programming Interface) is a **set of rules that allows applications to communicate with each other over HTTP**. It is based on the principles of REST, an architectural style defined by Roy Fielding in his doctoral dissertation.



(Coding Notes Bundle in bio)

Key Principles of REST

Stateless: Each request from a client to a server must contain all the information needed to understand and process the request. The server does not store any client context.

Client-Server Architecture: The client and server operate independently, promoting separation of concerns.

Cacheable: Responses must define themselves as cacheable or not, to improve performance.

Uniform Interface: REST APIs use standard HTTP methods (GET, POST, PUT, DELETE, etc.) and follow standard conventions for interacting with resources.

Layered System: The architecture can have layers, improving scalability and encapsulating logic.

Code on Demand (Optional): Servers can extend the functionality of clients by sending code (e.g., JavaScript) for execution.

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REST API Operations

HTTP Method	CRUD Operation	Description
GET	Read	Retrieve data from the server.
POST	Create	Send data to the server to create a resource.
PUT	Update	Update an existing resource.
DELETE	Delete	Remove a resource.
PATCH	Update (Partial)	Partially update a resource.

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REST API Components

Resources: The entities or data your API interacts with. Resources are represented as URLs.

- **Example:** /users, /products, /orders

Endpoints: The specific URLs where the API can be accessed.

- **Example:** https://api.example.com/v1/users

HTTP Status Codes:

- **200 OK:** Request succeeded.
- **201 Created:** Resource created successfully.
- **400 Bad Request:** Invalid input or request.
- **401 Unauthorized:** Authentication required.
- **404 Not Found:** Resource not found.
- **500 Internal Server Error:** Server-side error.

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